

LUMRA ERP

Page Architecture Blueprint

BAGIAN 1: MASTER DATA & SETUP

Disusun: 05 May 2026

DAFTAR ISI

1. PENDAHULUAN
2. HALAMAN KONFIGURASI & SETUP
 - Settings / Konfigurasi Umum
 - Setup Lokasi (Locations)
 - Setup Kategori Produk
 - Setup Unit Pengukuran
3. MASTER DATA - INVENTORY
 - Daftar Produk
 - Tambah/Edit Produk
 - Stok Awal / Adjustment
4. MASTER DATA - PENJUALAN
 - Daftar Customer
 - Tambah/Edit Customer
 - Daftar Vendor
 - Tambah/Edit Vendor
5. MASTER DATA - PRODUKSI
 - Daftar Recipe
 - Tambah/Edit Recipe
6. LAMPIRAN

1. PENDAHULUAN

1.1 Struktur Master Data

Master Data adalah fondasi sistem ERP. Data ini bersifat relatif statis namun sangat penting karena menjadi referensi untuk semua transaksi. Master Data LUMRA mencakup: Lokasi, Kategori, Unit, Produk, Customer, Vendor, dan Recipe.

1.2 Prinsip Desain Page Architecture

- 1. Form Progresif: Tampilkan field esensial terlebih dahulu
- 2. Validasi Real-time: Cek keunikan kode saat user mengetik
- 3. Master-Detail Pattern: Daftar di halaman utama, form terpisah
- 4. Bulk Operations: Sediakan opsi import CSV
- 5. Audit Trail: Simpan siapa yang membuat/edit dan kapan
- 6. Soft Delete: Gunakan flag `is_active` bukan delete fisik

2. HALAMAN KONFIGURASI & SETUP

2.1 Settings / Konfigurasi Umum

Nama Halaman & File Path	settings/general_settings.html
Konsep UI/UX	Tabs: Business Info, Fiscal Year, Currency. Form: Nama perusahaan, Alamat, NPWP, Logo
Logika View (Backend)	GET: Load settings dari LumraConfigBusiness. POST: Validate & update. Decorators: @login_required, @permissi
Koneksi Database	Model: LumraConfigBusiness. Read: company_name, address, currency, fiscal_year_start. Update: semua field
Alur Data	User -> Settings Page -> Submit Form -> Update DB -> Success Message -> Dashboard

2.2 Setup Lokasi (Locations)

Nama Halaman & File Path	settings/locations_list.html & location_form.html
Konsep UI/UX	List tabel: Kode, Nama, Tipe, Status. Actions: Edit, Delete (soft), Detail. Form modal untuk add/edit
Logika View	GET (list): pagination & filtering. GET (form): pre-fill. POST: validate code unique. DELETE: soft delete
Koneksi Database	Model: LumraConfigLocations. Read: id, code, name, location_type. Update: name, address, location_type
Alur Data	List -> Add -> Modal Form -> Submit -> Validate -> Save -> List Refresh

2.3 Setup Kategori Produk

File Path: settings/categories_list.html

UI/UX: Tree view hierarchical dengan expand/collapse

Backend: GET tree structure dengan parent-child. POST validate circular reference

Database: ProductCategories dengan FK self-referencing (parent_id)

Flow: Tree View -> Add/Edit -> Modal -> Validation -> Save -> Tree Refresh

2.4 Setup Unit Pengukuran

File Path: settings/units_list.html

UI/UX: Simple list: Code, Name, Abbreviation. Action buttons

Backend: Prevent delete jika sudah digunakan di products

Database: Units model. Check FK di ProductItems sebelum delete

Flow: List -> Add -> Form -> Validate Unique -> Save -> List

3. MASTER DATA - INVENTORY

3.1 Daftar Produk (Product List)

File Path: inventory/products_list.html

UI/UX: Tabel: SKU, Nama, Kategori, Unit, Status, Stok. Filter kategori & status. Search SKU/nama. Bulk actions. Pagination 25 items/page

Backend: GET dengan pagination, filtering, search. Prefetch Category & Unit. Hitung stok real-time dari InventoryTransaction. Cache 5 menit

Database: ProductItems (main), ProductCategories (FK), Units (FK). Aggregate stok dari InventoryTransaction

Flow: Dashboard -> Inventory -> Products -> List. User: View Detail, Edit, Duplicate, Delete (soft)

3.2 Tambah/Edit Produk (Product Form)

File Path: inventory/product_form.html

UI/UX: Tabs: Basic Info (Code, Name, Category, Unit, Description), Pricing (Buy/Sell price, Markup %), Stock (Initial stock, Min level, Reorder qty), Attributes

Backend: GET (add) form kosong + list kategori & unit. GET (edit) pre-fill. POST validate code unique. Insert/update ProductItems. Jika stok awal > 0 insert InventoryTransaction

Database: ProductItems, ProductCategories, Units. Write InventoryTransaction jika needed

Flow: Product List -> Add -> Form (Tabs) -> Fill Data -> Submit -> Validate -> Save -> Success -> List

3.3 Stok Awal / Adjustment

File Path: inventory/stock_adjustment.html

UI/UX: Select Lokasi. Tabel item: Product, Current Qty, New Qty, Difference. Add row button. Reason field (Opname, Rusak, Hilang). Preview total

Backend: GET form dengan list lokasi & current stok. POST validate qty > 0. Create InventoryTransaction per item. Create audit log

Database: InventoryTransaction (write), LumraConfigLocations (read)

Flow: Inventory -> Stock Adjustment -> Select Location -> Adjust Qty -> Submit -> Audit Log -> Success

4. MASTER DATA - PENJUALAN

4.1 Daftar Customer (Customer List)

File Path: sales/customers_list.html

UI/UX: Tabel: Kode, Nama, Email, Phone, City, Status. Filter status & city. Search nama/email. Action: View Profile, Edit, Delete, Deactivate. Metrics: Last Purchase Date, Total Value

Backend: GET customers + aggregate last purchase & total dari Orders. Prefetch LoyaltyStampCards

Database: LumraConfigCustomers (main), LoyaltyStampCards (FK). Aggregate: Last purchase & Total dari Orders

Flow: Sales -> Customers -> List -> Click Customer -> Profile / Edit / Purchase History

4.2 Tambah/Edit Customer (Customer Form)

File Path: sales/customer_form.html

UI/UX: Tabs: Personal Info (Code, Nama, Email, Phone, DOB), Address, Contact, Settings (Credit limit, Payment term, Loyalty tier)

Backend: GET (add) form + list loyalty tiers. GET (edit) pre-fill. POST validate email & code unique

Database: LumraConfigCustomers. Write: name, email, phone, address, city, credit_limit

Flow: Customer List -> Add -> Form (Tabs) -> Fill -> Submit -> Validate Unique -> Save -> List

4.3 Daftar Vendor (Vendor List)

File Path: purchasing/vendors_list.html

UI/UX: Tabel: Kode, Nama, Email, Phone, City, Status. Filter status & supplier category. Search. Metrics: Total Orders, Last Order Date, Payment Term

Backend: GET vendors + aggregate total orders & last order dari PurchaseOrders

Database: LumraConfigVendors (main), PurchaseOrders (FK)

Flow: Purchasing -> Vendors -> List -> Click -> Profile / Edit / Orders History

4.4 Tambah/Edit Vendor (Vendor Form)

File Path: purchasing/vendor_form.html

UI/UX: Tabs: Company Info (Code, Name, Tax ID, Website), Address, Contact, Supplier Categories (Checkboxes), Payment Terms

Backend: GET (add) form + list supplier categories. GET (edit) pre-fill + selected categories. POST validate code & tax_id unique. Save M2M categories

Database: LumraConfigVendors, VendorCategories (M2M)

Flow: Vendor List -> Add -> Form (Tabs) -> Fill -> Submit -> Validate -> Save -> List

5. MASTER DATA - PRODUKSI

5.1 Daftar Recipe (Recipe List)

File Path: production/recipes_list.html

UI/UX: Tabel: Kode, Nama, Output Product, Yield Qty, Status. Filter category & status. Search recipe name. Action: View, Edit, Duplicate, Archive. Tooltip: Bahan utama

Backend: GET recipes + prefetch output product. Show ingredient count

Database: Recipes (main), ProductItems (FK for output)

Flow: Production -> Recipes -> List -> Click -> View / Edit / Duplicate

5.2 Tambah/Edit Recipe (Recipe Form)

File Path: production/recipe_form.html

UI/UX: Tabs: Basic Info (Code, Name, Output Product, Yield Qty, Unit), Ingredients (Tabel dengan Product, Qty, Unit, Cost; Add row; Auto-calculate total), Instructions (WYSIWYG), Attributes

Backend: GET (add) form + product dropdown. GET (edit) pre-fill + ingredient rows. POST validate output product exists, yield qty > 0. Save Recipes + RecipeIngredients

Database: Recipes, RecipeIngredients (FK to Recipes & ProductItems)

Flow: Recipe List -> Add -> Form (Tabs) -> Add Ingredients -> Submit -> Validate -> Save

6. LAMPIRAN

6.1 Standard View Function Contract

- LIST VIEW: GET dengan pagination, filtering, searching, prefetch relations
- FORM VIEW: GET (render form kosong/prefill), POST (validate & save)
- DELETE VIEW: Soft delete (set is_active=False), bukan delete fisik
- DECORATORS: @login_required, @permission_required sesuai keperluan
- ERROR HANDLING: Try-catch untuk database errors, return meaningful message
- AUDIT LOG: Log siapa, apa, kapan untuk setiap create/update/delete

6.2 Form Validation Checklist

- ✓ Unique constraint: Code, Email, Tax ID (check DB sebelum save)
- ✓ Required fields: Semua field wajib terisi (form validation)
- ✓ Format validation: Email (regex), Phone (11-12 digits), Date (ISO)
- ✓ Business logic: Code pattern (PREFIX-NNNN), Credit limit > 0
- ✓ Foreign key: Parent exists sebelum save child
- ✓ Circular reference: Jika hierarchical, cek tidak ada self-reference
- ✓ Soft delete: Set is_active=False, jangan delete fisik
- ✓ Audit log: Simpan who, what, when untuk perubahan